

Shock Problem and PINN Framework

1 Problem Setup

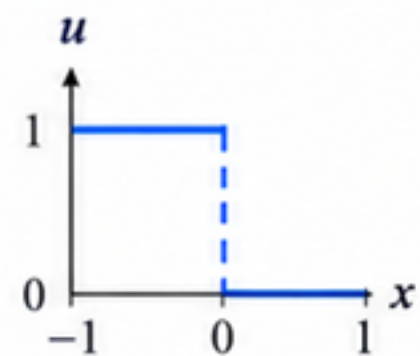
Consider the 1D Burgers Riemann problem

Governing equation $u_t + uu_x = 0, \quad x \in [-1, 1], \quad t \in (0, 1]$

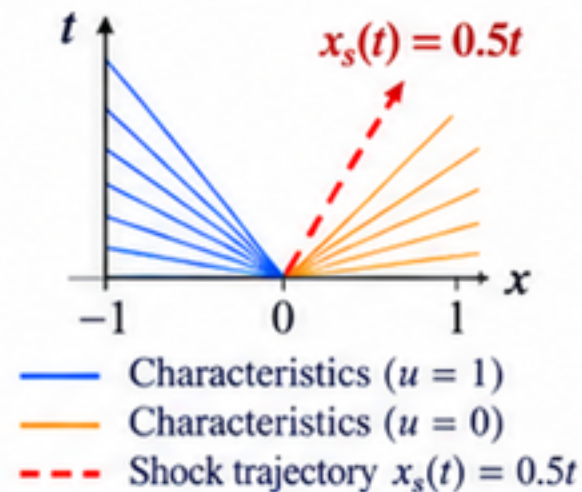
Initial condition $u(x, 0) = \begin{cases} 1, & x < 0, \\ 0, & x > 0. \end{cases}$

Boundary condition $u(-1, t) = 1, \quad u(1, t) = 0, \quad t \in (0, 1]$

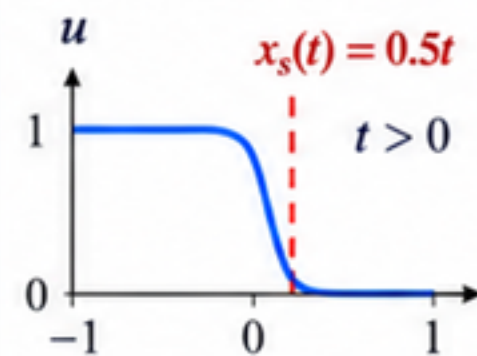
(a) Initial condition



(b) Characteristics and shock trajectory



(c) Solution profile for $t > 0$



- Wave speed is determined by u
- The left high- u region propagates faster
- Characteristic lines merge and form a shock

2 PINN Framework

